

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 2

SUBTRACTION OF TWO 8-BIT NUMBERS

AIM: To write and execute an 8085 assembly language program to subtract two 8-bit numbers and store the result in memory.

ALGORITHM

1. Load the first number from 2500H.
2. Move it to register B.
3. Load the second number from 2501H.
4. Subtract register B from the accumulator.
5. Store the result in 2502H.
6. Stop the program.
- 7.

CODE:

```
LDA 2500H
MOV B,A
LDA 2501H
SUB B
STA 2502H
HLT
```

OUTPUT:

Registers

A	01
BC	05 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 0C
SP	FF FF
Int-Reg	00

Flag

S	0
Z	0
AC	0
P	0
C	0

Decimal - Hex Conversion

Decimal

Hex

0

0

To Hex

To Dec

I/O Ports

0

-

+

00

Update Port Value

Memory

8001

-

+

06

Update Memory

Load me at

1

LDA

8000

2

MOV

B,A

3

LDA

8001

4

SUB

B

5

SXA

8002

6

HLT

7

Memory

Start 8000

OK

Address (Hex)	Address	Data
1F40	8000	5
1F41	8001	6
1F42	8002	1
1F43	8003	0
1F44	8004	0
1F45	8005	0
1F46	8006	0
1F47	8007	0
1F48	8008	0
1F49	8009	0
1F4A	8010	0
1F4B	8011	0
1F4C	8012	0
1F4D	8013	0

Line No

Assembler Message

0

Program assembled successfully

Simulator: Idle

Search

ENG IN

13:52 08-08-2026